








Advanced Time Series Analysis and Forecasting

Chapter 13: Foundation models and conformal prediction



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Acronyms used in this material (1/2)

- **ACI**: Adaptive Conformal Inference
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **API**: Application Programming Interface
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARX**: AutoRegressive model with eXogenous variables
- **BET**: Bucharest Exchange Trading (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BOOM**: Benchmark Of Observability Metrics
- **CEE**: Central and Eastern Europe
- **CPM**: Contiguous Patch Masking
- **CPU**: Central Processing Unit
- **CQR**: Conformalized Quantile Regression
- **CRPS**: Continuous Ranked Probability Score
- **DM**: Diebold–Mariano (test)
- **DQ**: Dynamic Quantile (test of Engle and Manganelli)
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **EODHD**: EOD Historical Data (market-data provider)
- **ETS**: Error, Trend, Seasonality (exponential smoothing state space models)
- **EU**: European Union
- **FDR**: False Discovery Rate
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GIFT**: General Time Series Forecasting Model Evaluation (the GIFT-Eval benchmark)
- **GPT-2**: Generative Pre-trained Transformer 2 (OpenAI language model)

Acronyms used in this material (2/2)

- **GPT4TS**: GPT for Time Series (One Fits All)
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **ID**: Identifier (arXiv or DOI)
- **LLM**: Large Language Model
- **LOTSA**: Large-scale Open Time Series Archive
- **MAE**: Mean Absolute Error
- **MASE**: Mean Absolute Scaled Error
- **MCS**: Model Confidence Set
- **MOMENT**: MOMENT, a family of open time-series foundation models
- **MSE**: Mean Squared Error
- **NVIDIA**: NVIDIA Corporation (graphics processors)
- **OOD**: Out Of Distribution
- **PID**: Proportional–Integral–Derivative (controller)
- **QLIKE**: Quasi-LIKElihood loss
- **RLHF**: Reinforcement Learning from Human Feedback
- **RV**: Realised Variance
- **TS**: Time Series (in TS-Arena)
- **TSA**: Time Series Analysis
- **VAT**: Value Added Tax
- **WQL**: Weighted Quantile Loss

Today's question and route

- **Question**: can one pretrained model forecast series it has never seen, and how do we turn any forecast, pretrained or not, into intervals whose coverage we can guarantee or at least control?
  - two halves of one problem: point and quantile accuracy (foundation models), then honest uncertainty (conformal prediction)
- **Route** of the chapter
  - pretraining: data, tokens, patches, scaling laws; model families and design choices
  - benchmarks, contamination and testing across many series; evidence on Romanian and EU data; LLMs and their critique
  - conformal prediction: exchangeability, split conformal, CQR, the limits of conditional coverage
  - dependent data: weights, EnbPI, ACI, conformal PID; diagnostics; calibrating foundation-model intervals and VaR
- Prerequisites
  - Seminar 13; TSA, Chapter 11 (slide Prior knowledge from TSA and new material)
  - Chapter 1 (scoring rules, DM, MCS), Chapter 9 (VaR backtests, ACI for VaR), Chapter 12 (deep architectures)

Learning outcomes

- Explain how a time series foundation model is pretrained: corpus, scaling, tokenisation or patching, output head and loss
- Compare model families and choose between zero-shot use, fine-tuning and in-context covariates
- Design a benchmark without leakage or contamination and test differences across many series with corrections for multiplicity
- Prove the coverage of split conformal and CQR, and state what conformal prediction cannot guarantee
- Apply weighted conformal, EnbPI, ACI and conformal PID to dependent data, diagnose coverage and calibrate foundation-model intervals

Reading, data and tools

- Foundation models and benchmarks
  - models: Ansari et al. (2024); Ansari et al. (2025); Das et al. (2024); Woo et al. (2024); Auer et al. (2025)
  - benchmarks: Aksu et al. (2024); Shchur et al. (2025); critique and evaluation: Tan et al. (2024); Hewamalage, Ackermann and Bergmeir (2023)
- Conformal prediction
  - theory: Vovk, Gammerman and Shafer (2022); Angelopoulos and Bates (2023); Romano, Patterson and Candès (2019); Barber et al. (2023)
  - time series: Gibbs and Candès (2021); Angelopoulos, Candès and Tibshirani (2023); Xu and Xie (2021); survey of forecasting practice: Petropoulos et al. (2022)
- Python Quantlets of this chapter: Quantlets/Ch_13
  - open checkpoints on a CPU: `chronos-forecasting` (Chronos-Bolt, Chronos-2), `timesfm` (TimesFM 2.5), `tirex-ts` (TiRex); conformal methods, DM, MCS and backtests written in `numpy`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Sample
Romanian electricity load, hourly	Energy-Charts (ENTSO-E transparency data)	2022 – 30 September 2026
Bucharest air temperature, hourly	Open-Meteo historical archive	2022 – 2026
HICP annual inflation, 27 EU countries	Eurostat, monthly	2000 – August 2026
Realised variance, 5-minute returns	Bitcoin: Binance; S&P 500: Oxford-Man Heber et al. (2009)	2018 – 2026; 2000 – 2022
BET, S&P 500, NVIDIA, daily	EODHD daily closes	2000 – 2026

- Returns are 100 times log differences; the realised variance is in squared percent; every forecast uses only data available at its origin

Origins of conformal prediction and foundation models



Royal Holloway, University of London: the Founder's Building

Photo: Diliff (2015); CC BY-SA 3.0; Wikimedia Commons

- 1990s–2005: conformal prediction at Royal Holloway: Vovk, Gammerman and Shafer, rooted in Kolmogorov's algorithmic randomness Vovk, Gammerman and Shafer (2005)
- 2002–2018: split (inductive) conformal Papadopoulos et al. (2002); distribution-free regression Lei et al. (2018); 2019: CQR Romano, Patterson and Candès (2019)
- 2021–2023: time series: ACI Gibbs and Candès (2021), EnbPI Xu and Xie (2021), conformal PID Angelopoulos, Candès and Tibshirani (2023), beyond exchangeability Barber et al. (2023)
- 2017–2025: Transformers Vaswani et al. (2017), foundation models Bommasani et al. (2021); for series: Lag-Llama, TimesFM, Moirai, Chronos (2023–2024), TiRex, Chronos-2 (2025)

Prior knowledge from TSA and new material

- Known (TSA, Chapter 11)
  - zero-shot use of Chronos, TimesFM, Moirai and Lag-Llama; mean scaling and quantisation in brief
  - pinball loss, CRPS, WQL, MASE; first warnings about contamination
- New: the research toolkit
  - design choices and their consequences (range limits, quantile heads, patching, covariates in context), scaling evidence
  - benchmarks with pre-registered windows after the model releases, tests across many series with Holm and BH corrections
  - conformal theory with proofs, its failure under dependence and the online methods that repair it, with coverage diagnostics
- Case studies: the designs of the papers, on our data
  - foundation models: Ansari et al. (2024); Tan et al. (2024)
  - conformal prediction: Romano, Patterson and Candès (2019); Gibbs and Candès (2021); Barber et al. (2023); Angelopoulos, Candès and Tibshirani (2023); Xu and Xie (2021)

A foundation model as an amortised forecaster (1/2)

- **Pretraining**: one network is fitted once, by minimising the forecast loss summed over all series and all origins of a large corpus

$$\hat{\theta} = \arg\min_{\theta} \sum_{s \in \mathcal{D}_{\text{pre}}} \sum_{t} \ell(y^{(s)}_{t+\mathbf{1}:t+H},\ f_\theta(y^{(s)}_{t-C+\mathbf{1}:t}))$$

- Notation
  - $\mathcal{D}_{\text{pre}}$: the pretraining corpus, a collection of many series; s indexes the series, t the forecast origin
  - $y^{(s)}_{t-C+\mathbf{1}:t}$: the last C values of series s (the **context**); $y^{(s)}_{t+\mathbf{1}:t+H}$: its next H values (the **horizon**)
  - f_θ: the network with weights θ; it maps a context to a forecast of the next H values (a distribution or a set of quantiles)
  - ℓ: a scoring rule (cross-entropy, pinball, negative log-likelihood); $\hat{\theta}$: the estimated weights
- The term **foundation model** Bommasani et al. (2021): trained once on broad data, then adapted to many tasks

A foundation model as an amortised forecaster (2/2)

- **Zero-shot** use: the weights stay at $\hat{\theta}$ and the forecast for a new series $y_1, \ldots, y_T$ is the network applied to its last C values

$$\hat{y}_{T+1:T+H} = f_{\hat{\theta}}(y_{T-C+1:T})$$

  - T: the last observed period of the target series; no parameter is estimated on it
- **Amortised inference**
  - a classical model is re-estimated on every new series; here the network performs the estimation step inside its forward pass
  - the cost of estimation is paid once, during pretraining, and shared by every later forecast
- Global model and foundation model
  - a global model of Chapter 12 is trained on the series of one data set
  - a foundation model is trained on many data sets, frequencies and domains

Structure shared across series

- Shapes recur across domains: seasonal profiles, damped trends, level shifts, volatility clusters, intermittency
  - after scaling, a load curve and a web-traffic curve can look alike: the model learns a prior over shapes
- Not shared: the economics of a particular series (a tax change, a policy rule, a holiday calendar)
  - unless it is passed as a covariate in the context (Chronos-2) or learned by fine-tuning
- A no-free-lunch caveat: averaged over all processes no forecaster wins; pretraining helps only if the target resembles the corpus
  - financial returns are close to a martingale difference: little shape to transfer, much to overfit

Pretraining data: scale and composition

- Chronos Ansari et al. (2024)
  - public data sets plus two augmentations
  - **TSMixup**: convex mixtures of real series; **KernelSynth**: series drawn from Gaussian processes with random composite kernels
- TimesFM Das et al. (2024); Moirai Woo et al. (2024)
  - TimesFM: about 10^{11} time points: Google Trends, Wikipedia page views, synthetic and public series
  - Moirai: LOTSA, over 27 billion observations in nine domains
- Toto Cohen et al. (2025); GIFT-Eval Aksu et al. (2024)
  - Toto: observability metrics of a cloud provider, a corpus 4–10 times larger than those of earlier models
  - GIFT-Eval: a **non-leaking** pretraining set of about 230 billion points
- Composition matters more than size for us: economic and financial series are a small, low-frequency share of every corpus

Tokenisation: from real values to a vocabulary (1/2)

- Chronos Ansari et al. (2024), step 1, **mean scaling**: every value of the context is divided by the mean absolute value of the context

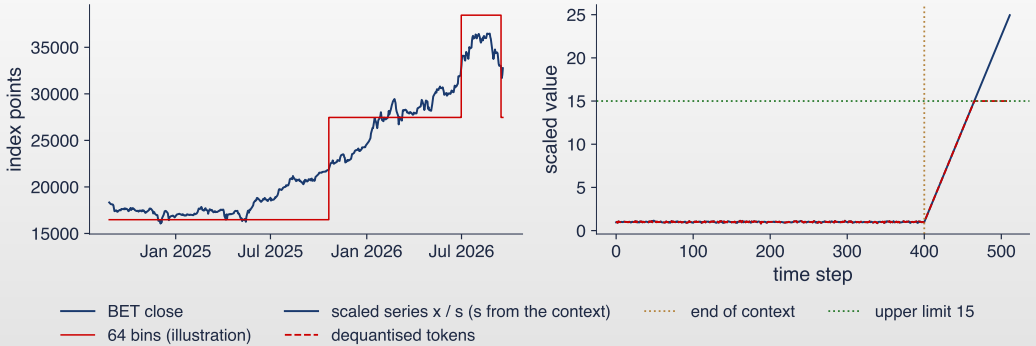
$$\tilde{x}_t = x_t / s, \qquad s = \frac{1}{C} \sum_{t=1}^{C} |x_t|$$

  - $x_1, \ldots, x_C$: the context values; s: the scale, computed on the context only; $\tilde{x}_t$: the scaled value, close to 1 in absolute value on average
- Step 2, **uniform bins**: the interval $[-15, 15]$ is covered by equally spaced bins, one token per bin; with the special tokens the vocabulary has 4096 tokens
  - each scaled value is replaced by the token of the bin that contains it
- Step 3: a language model (T5) trained with **cross-entropy** predicts the next token
  - the forecast is a categorical distribution over bins, sampled autoregressively (one token at a time)
  - no notion of distance between tokens: neighbouring bins are as different as distant ones, unless the data teach otherwise

Tokenisation: from real values to a vocabulary (2/2)

- **Resolution**: neighbouring bin centres are $30/4092$ apart on the scaled axis, i.e. $30s/4092$ in the units of the series
  - $30 = 15 - (-15)$ is the length of the covered range; the quantisation error is at most half of this spacing
- **Hard range**: a value above $15s$ (or below $-15s$) has no token and cannot be predicted
- **Scale invariance**: multiplying the series by $c > 0$ multiplies s by c and leaves the tokens unchanged
- Other input representations
  - LLMTime Gruver et al. (2023): the digits of each value as text tokens
  - Lag-Llama Rasul et al. (2023): lagged values as features
  - patch models: no vocabulary at all (next slides)

The Chronos tokeniser on real data



- Left: BET closes, 27 August 2024 – 18 September 2026 ($C = 512$), and a 64-bin quantisation (illustration); right: a context near 1 followed by a rise to 25 times that level

ATS_ch13_pretraining

Interpreting the tokeniser

- BET: $s = 23\,072$ points, bin width 169.1 points, maximum quantisation error 84.5 points: negligible for prices, but a fixed share of the level
- Right panel: the scale is set by the context, so the path that rises beyond $15s$ is clipped: 42% of the future steps are unreachable, the last one by 40%
- Explosive series (bubbles, Chapter 16; hyperinflation) are outside the support of a mean-scaled vocabulary; Chronos-2 replaces it by an arcsinh transform and quantile outputs

Patching and the output head

- **Patching** Nie et al. (2023): the context is split into patches of P consecutive values and each patch becomes one input token
  - P: the patch length; a context of C values gives C/P tokens
  - attention compares every pair of tokens, so its cost grows as $O((C/P)^2)$ instead of $O(C^2)$; $O(\cdot)$: order of magnitude
  - TimesFM Das et al. (2024): input patches of 32, output patches of 128 (fewer autoregressive steps); Chronos-Bolt and Chronos-2: patches of 16; Moirai Woo et al. (2024): several sizes by frequency
- Output heads
  - categorical over bins (Chronos)
  - **quantile head** trained by the pinball loss (Chronos-Bolt: 10%–90%; Chronos-2: 1%–99%; TimesFM 2.5, TiRex: deciles)
  - parametric: Student-t (Lag-Llama Rasul et al. (2023)), a mixture of distributions (Moirai)
- **Direct multi-step quantiles**: one pass gives the quantiles of all H steps
  - no accumulation of errors from step to step
  - but the output is the marginal distribution of each step, not their joint distribution: a path functional (a maximum, a sum over the horizon) cannot be read from it

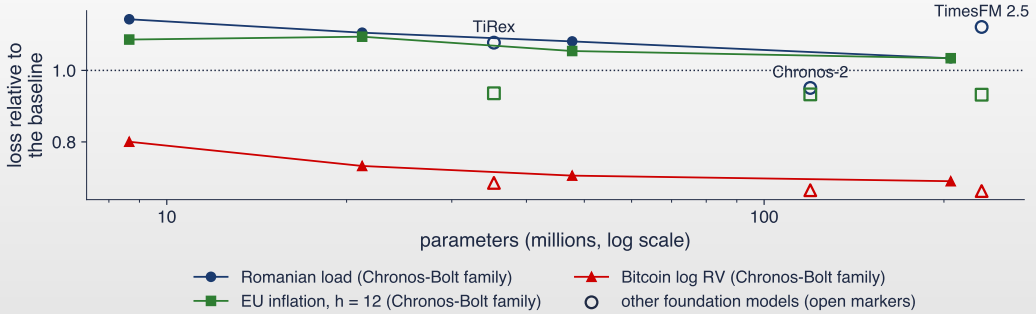
Scaling laws

- Language models Kaplan et al. (2020), Hoffmann et al. (2022): the test loss falls as a power law of the model size, and similarly of the data size and of the compute, over several orders of magnitude

$$L(N) \approx (N_c/N)^{\alpha_N}$$

  - $L(N)$: test loss of a model with N parameters; N_c: a fitted scale constant; $\alpha_N > 0$: the fitted exponent
  - reading: doubling N multiplies the loss by $2^{-\alpha_N}$; on a log–log plot the relation is a straight line of slope $-\alpha_N$
- Time series: decoder-only Transformers show the same power laws in parameters, data and compute Edwards et al. (2024); the look-back length interacts with data size Shi et al. (2024)
  - out-of-distribution: the log-likelihood scales similarly in and out of distribution, but architecture matters; tweaks that help in distribution can reduce OOD scalability Yao et al. (2025)
- A scaling law concerns the average loss on the corpus distribution
  - it says nothing about one particular Romanian series
  - small models (TiRex, 35 million parameters) beat larger ones on public leaderboards Auer et al. (2025)

Size and accuracy on our three tasks



- Losses relative to the baseline of each task: Romanian load (MAE / expert ARX, 2025–2026), EU inflation (MAE / random walk, $h = 12$, geometric mean over 27 countries), Bitcoin log RV (MSE / HAR)
- Parameters counted in the loaded checkpoints; a value below 1 means the model beats the baseline

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Interpreting size against accuracy

- Chronos-Bolt family (9–205 M parameters): load 1.143 → 1.034, inflation 1.086 → 1.034, Bitcoin 0.800 → 0.690
- Across families size is not the ranking: Chronos-2 (119 M) 0.951 on load; TiRex (35 M) 1.078; TimesFM 2.5 (231 M) 1.121
- Within the Chronos-Bolt family the loss falls with size on all three tasks (one exception: mini on inflation)
  - across families, architecture, corpus and training procedure change together
  - so three tasks are not a scaling study

Recap: pretraining

- Pretraining amortises estimation over a corpus; zero-shot use runs the estimator inside one forward pass
- Mean scaling and binning give scale invariance but a hard range; patching and quantile heads trade it for direct multi-step quantiles
- Scaling laws hold on average over corpora; on a given economic series, design and data matter more than size

The main open models

Model	Architecture	Parameters	Output
Chronos Ansari et al. (2024)	T5 encoder–decoder on tokens	20–710 M	categorical, sampled paths
Chronos-Bolt	T5 on patches of 16	9–205 M	9 quantiles, 10%–90%, 64 steps
Chronos-2 Ansari et al. (2025)	encoder with group attention	119 M	21 quantiles, 1%–99%; covariates
TimesFM 2.5 Das et al. (2024)	decoder-only, patches	231 M	mean and deciles
Moirai Woo et al. (2024)	masked encoder, any-variate attention	14–311 M	mixture distribution
Lag-Llama Rasul et al. (2023)	decoder-only on lag features	small	Student-t
MOMENT Goswami et al. (2024)	masked T5 encoder, multi-task	40–385 M	reconstruction; heads per task
TiRex Auer et al. (2025)	xLSTM	35 M	deciles

- Toto Cohen et al. (2025) (151 M, observability data) and Moirai 2.0 Liu et al. (2025) are open as well; TimeGPT Garza et al. (2023) is reached only through a paid API

The Chronos family

- **Chronos** Ansari et al. (2024)
  - the training procedure of language models, unchanged; forecasts by sampling token paths
  - zero-shot results comparable to models trained on the target data (their Benchmark II)
- **Chronos-Bolt**
  - patch inputs and a direct quantile decoder: much faster than Chronos
  - context up to 2048, 64 steps; quantiles only between 10% and 90%
  - a 95% interval or a VaR 1% cannot be read from its output: the request is clipped to the nearest trained level
- **Chronos-2** Ansari et al. (2025)
  - group attention shares information across the series of a group (variates, related series, covariates)
  - **in-context learning** of covariate effects, without re-estimation
  - trained largely on synthetic multivariate structures imposed on univariate series
  - context 8192; 21 quantiles from 1% to 99%; arcsinh scaling

TimesFM, Moirai, Lag-Llama, MOMENT (1/2)

- **TimesFM** Das et al. (2024)
  - decoder-only, with patches; the output patch is longer than the input patch
  - trained on many granularities; version 2.5 adds a quantile head and long contexts
- **Moirai** Woo et al. (2024)
  - masked encoder; frequency-specific patch sizes
  - any-variate attention flattens multivariate inputs into one sequence
  - mixture output: Student-t, log-normal, negative binomial
  - Moirai 2.0 Liu et al. (2025): a decoder-only simplification, smaller and better on GIFT-Eval

TimesFM, Moirai, Lag-Llama, MOMENT (2/2)

- **Lag-Llama** Rasul et al. (2023)
  - a LLaMA-type decoder whose tokens are vectors of lagged values at many seasonal lags
  - Student-t output head; strong after fine-tuning
- **MOMENT** Goswami et al. (2024)
  - pretrained by masked reconstruction on the Time series Pile
  - one encoder for forecasting, classification, anomaly detection and imputation

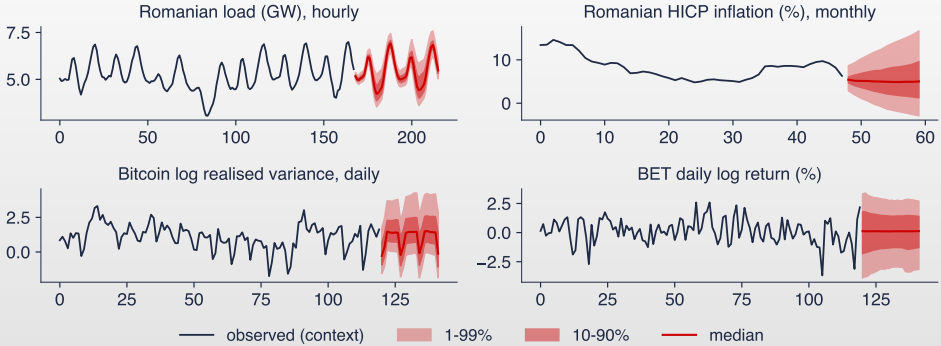
TiRex, Toto and TimeGPT

- **TiRex** Auer et al. (2025)
  - an xLSTM keeps a state across the context, which allows state tracking over long horizons
  - xLSTM: the recurrent cells of Chapter 12 with exponential gating and a matrix memory
  - contiguous patch masking (CPM) in training: the model learns to forecast several steps without feedback
- **Toto** Cohen et al. (2025)
  - decoder-only, for multivariate observability metrics; the BOOM benchmark (2807 series)
  - the domain composition of the corpus drives the results
- **TimeGPT** Garza et al. (2023)
  - closed weights and an undisclosed corpus, behind a paid API
  - contamination cannot be audited; results are not reproducible if the service changes the model
  - the data leave your institution

Zero-shot, fine-tuning, in-context covariates

- **Zero-shot**: the forecast is $f_{\hat{\theta}}(y_{T-C+1:T})$
  - no training, hence no risk of overfitting the target; the only choices are C and the model
- **Fine-tuning**: the weights start at $\hat{\theta}$ and take a few gradient steps on the target data
  - all weights, or low-rank adapters (LoRA) added to frozen weights
  - needs a validation split respecting time (Chapter 12); can forget the prior; with short economic series it rarely pays
- **In-context covariates** (Chronos-2)
  - past values of a covariate x_t (e.g. temperature) and future values of regressors known in advance enter the context
  - the model infers their effect inside the forward pass, with no parameter estimated
  - cross-learning: forecasting a batch of related series jointly (all 27 EU countries at one origin)
- Known future covariates must really be known at the origin
  - weather forecasts, not realised weather, unless an upper bound is the goal

Zero-shot forecasts of Chronos-2 on four data sets



- Chronos-2 from the last origin of each data set: Romanian load (48 hours), Romanian HICP inflation (12 months), Bitcoin log realised variance and BET daily returns (22 days); bands 10–90% and 1–99%

ATS_ch13_zero_shot

Interpreting the four forecasts

- Load: the daily and weekly shapes are continued with narrow bands: the most favourable case for shape transfer
- Inflation: a smooth path towards the recent level with bands that widen with the horizon: close to what an AR model with persistence would give
- Log RV of Bitcoin: a weekly pattern (lower variance at weekends) around a level set by the context
- Returns: a flat median and a symmetric band, i.e. an unconditional distribution
- Return bands from the 1% and 99% quantiles: Chronos-2 VaR 1% for the next day is 3.90 (% of value); Section 8 checks whether such numbers are calibrated

Recap: model families

- Families differ in input (tokens, patches, lags), architecture (encoder, decoder, recurrent) and output (categorical, quantiles, parametric)
- Quantile range is a design constraint: deciles only, except Chronos-2
- Zero-shot is the default; covariates in context and cross-learning are the new levers; closed APIs cannot be audited

Benchmarks for pretrained models

- **GIFT-Eval** Aksu et al. (2024): 23 data sets, over 144 000 series, 177 million points, seven domains, ten frequencies, short to long horizons; a non-leaking pretraining set
- **fev-bench** Shchur et al. (2025): 100 tasks in seven domains, 46 with covariates; win rates and skill scores with bootstrap confidence intervals
- **Live benchmarks**: forecasts registered before the outcomes exist, e.g. TS-Arena Meyer et al. (2025) and Impermanent Garza et al. (2026): the only design immune to contamination by construction
- Classical references for what a fair comparison needs: M5 Makridakis, Spiliotis and Assimakopoulos (2022); pitfalls catalogued by Hewamalage, Ackermann and Bergmeir (2023)

Aggregating scores across series (1/2)

- **MASE** Hyndman and Koehler (2006): the mean absolute forecast error divided by the in-sample mean absolute error of the seasonal naive forecast

$$\mathrm{MASE} = \frac{\frac{1}{H}\sum_{h=1}^{H} |y_{T+h} - \hat{y}_{T+h}|}{\frac{1}{T-m}\sum_{t=m+1}^{T} |y_t - y_{t-m}|}$$

  - $\hat{y}_{T+h}$: the forecast h steps after the origin T; m: the seasonal period (1 without seasonality)
  - MASE < 1: better than the naive forecast in sample; free of the units of y

- **WQL** (weighted quantile loss): the pinball losses of the forecast quantiles, summed and divided by the total absolute level of the series

$$\mathrm{WQL} = \frac{2\sum_t \sum_\tau \rho_\tau(y_t - \hat{q}_{\tau,t})}{\sum_t |y_t|}, \qquad \rho_\tau(u) = u(\tau - 1\{u < 0\})$$

  - $\hat{q}_{\tau,t}$: the forecast quantile of level $\tau \in (0,1)$ for period t; $1\{\cdot\}$: 1 if the condition holds, 0 otherwise
  - ρ_τ: the pinball loss; its average over τ approximates the CRPS (Chapter 1); WQL $= 0$ only for perfect quantiles

Aggregating scores across series (2/2)

- **Relative score** of series s: the loss of the model divided by the loss of a baseline on the same series

$$r_s = \frac{L_s(\text{model})}{L_s(\text{baseline})}, \qquad \bar{r}_{\text{geo}} = \Big(\prod_{s=1}^{S} r_s\Big)^{1/S}$$

  - L_s: a loss (MAE, MASE, WQL) on series s; S: the number of series; $r_s < 1$: the model beats the baseline on series s
- **Geometric mean** $\bar{r}_{\text{geo}}$ Fleming and Wallace (1986): the rankings do not depend on the choice of baseline, and a gain and a loss of the same ratio cancel
  - **skill score** $1 - \bar{r}_{\text{geo}}$: the average proportional gain over the baseline (0: no gain)
  - **win rate**: the share of series on which a model beats another
- Two aggregations to avoid
  - the arithmetic mean of ratios: dominated by the series on which the baseline is weak
  - the mean of raw errors: dominated by the series with the largest scale

Leakage and contamination

- **Series leakage**
  - an evaluation series (or a near copy) is in the pretraining corpus
  - Chronos separates in-domain (Benchmark I) from zero-shot (Benchmark II) results for this reason Ansari et al. (2024)
- **Temporal contamination**
  - the test window precedes the training cutoff: the model may have seen the outcomes of other, correlated series in the same period
  - a two-data-set design against contamination, for electricity prices Pan and Ezzat (2026)
- Designs that help
  - evaluate only after the release (or the declared cutoff) of every model in the comparison; we use 1 November 2025, after the last of our models (Chronos-2, 30 October 2025)
  - compare the relative skill before and after: a large drop after the release is a warning sign
  - pre-register origins, horizons, metrics and baselines before looking at the results

Testing across many series (1/2)

- One series (all in Chapter 1)
  - DM with HLN correction Diebold and Mariano (1995), Harvey, Leybourne and Newbold (1997); conditional ability Giacomini and White (2006)
  - many models: MCS Hansen, Lunde and Nason (2011)
- S series tested separately, each at level 5%: under the null hypothesis about $0.05S$ false rejections are expected
  - with $S = 27$ countries, about one or two "significant countries by chance alone
- Two error rates for a family of tests
  - **FWER** (family-wise error rate): the probability of at least one false rejection
  - **FDR** (false discovery rate): the expected share of false rejections among all rejections
- Notation: $p_{(1)} \leq \cdots \leq p_{(S)}$ are the S p-values sorted increasingly; α: the target level of the family (e.g. 5%)

Testing across many series (2/2)

- **Holm** Holm (1979), controls the FWER: go up the sorted list and reject while

$$p_{(k)} \leq \frac{\alpha}{S-k+1}, \qquad k = 1, 2, \ldots$$

  - the smallest p-value faces the Bonferroni threshold α/S, the next ones gradually looser thresholds; stop at the first failure
- **Benjamini–Hochberg** Benjamini and Hochberg (1995), controls the FDR: reject the k^* smallest p-values, with

$$k^* = \max\{k : p_{(k)} \leq k\alpha/S\}$$

  - more rejections than Holm; valid under independence or positive dependence between the tests
- **Pooled test**: average the loss differential across series at each origin, then run one DM test with HAC variance
  - the correlation between series stays in the averaged time series, so the HAC variance accounts for it
- Ranks over many data sets: Friedman and Nemenyi tests Demšar (2006); they ignore the size of differences and assume independent data sets

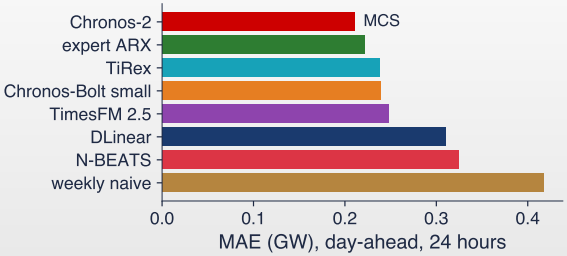
Recap: benchmarks and testing

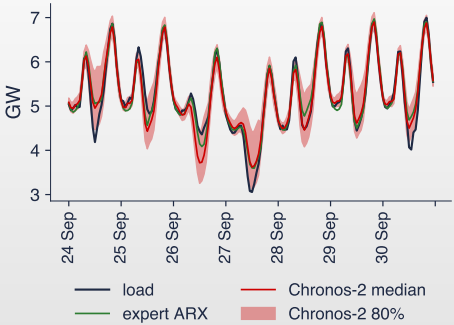
- Aggregate scale-free relative scores by geometric means and report uncertainty over series
- Contamination is a property of the pair (model, test window): evaluate after the release or live
- Many series mean many tests: correct with Holm or BH, or pool before testing

The design of our comparisons

- **Romanian load**: day-ahead, 24 hours, every day from 1 January 2025 to 30 September 2026 (638 days); context 2048 hours for the foundation models
  - baselines: weekly naive; the expert ARX of Ziel and Weron (2018) for each hour (Chapter 1); DLinear Zeng et al. (2023) and N-BEATS Oreshkin et al. (2020) trained globally (Chapter 12)
- **EU inflation**: 27 countries, rolling monthly origins January 2012 – August 2025, $h = 1, \ldots, 12$; random walk, AR(p) by AIC, damped ETS Hyndman et al. (2002), DLinear and N-BEATS trained on the panel
- **Realised variance**: one day ahead, log RV; HAR Corsi (2009) on a rolling 1000-day window; QLIKE Patton (2011)
- Foundation models: Chronos-Bolt small, Chronos-2, TimesFM 2.5, TiRex, zero-shot, on a CPU; Moirai and Lag-Llama were not run (their libraries need older software versions)

Romanian load: day-ahead accuracy





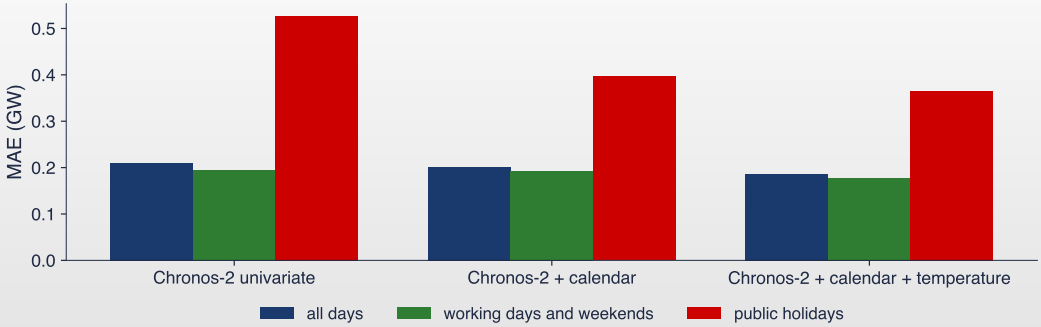
- Left: MAE (GW) over 24 hours and 638 days, models in the 10% MCS marked; right: the last week of the sample, expert ARX and Chronos-2 with its 80% band

ATS_ch13_zero_shot

Interpreting the load comparison

- MAE: Chronos-2 0.210, expert ARX 0.221, TiRex 0.238, Chronos-Bolt 0.239, TimesFM 0.248; DLinear 0.311, N-BEATS 0.325, weekly naive 0.417 GW
- DM against the expert ARX: Chronos-2 $t = -3.05$ ($p = 0.002$); the other three foundation models are significantly worse; the 10% MCS is {Chronos-2}
- 80% interval coverage: Chronos-2 75.5%, Chronos-Bolt 81.2%, TimesFM 81.2%, expert ARX 78.1%: none is far off, none is exact
- A univariate zero-shot model matches an established domain model on a series with strong, regular shapes; the global deep models trained on three years lose to both

In-context covariates: calendar and temperature



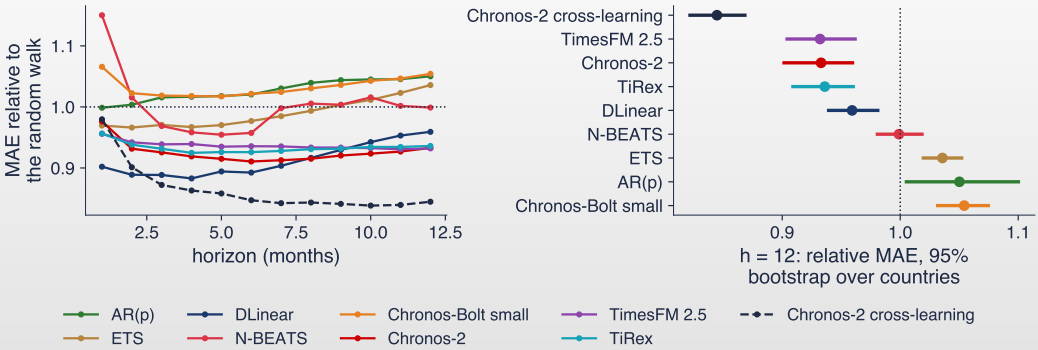
- Chronos-2 MAE on all days, on ordinary days and on the 29 public holidays: univariate, with weekend and holiday flags, and with flags plus Bucharest hourly temperature (realised values for the forecast day: an upper bound)

ATS_ch13_zero_shot

Interpreting the covariate experiment

- Holidays: MAE 0.527 univariate, 0.396 with the calendar, 0.365 with temperature; ordinary days: 0.195, 0.193, 0.178
- Calendar against univariate: DM $t = -2.79$ ($p = 0.005$); temperature on top: $t = -7.02$ ($p < 0.001$)
- The holiday effect is learned in context from a few earlier holidays in the 2048-hour window; no parameter was estimated
- Realised temperature is not available at the origin: an operational test needs archived weather forecasts

EU inflation: 27 countries, 12 horizons



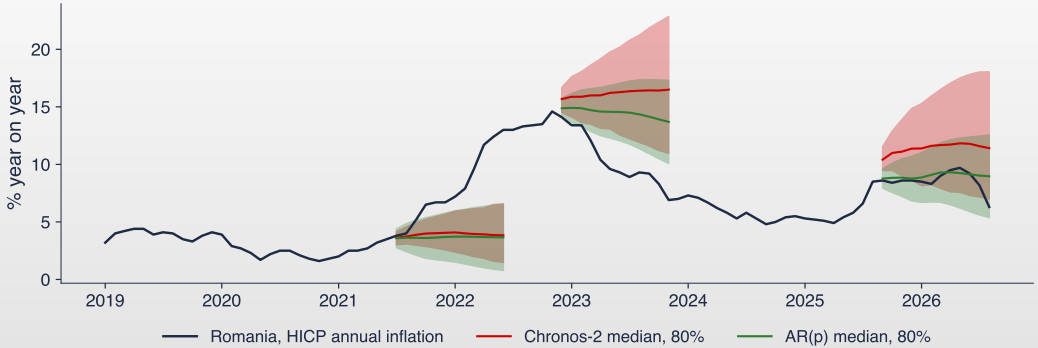
- Left: MAE relative to the random walk by horizon, geometric mean over countries; right: $h = 12$ with 95% bootstrap intervals over countries; origins January 2012 – August 2025

ATS_ch13_benchmark

Interpreting the inflation panel

- $h = 1$: AR 0.999, Chronos-2 0.976, TimesFM 0.956, TiRex 0.956; $h = 12$: AR 1.050, ETS 1.036, Chronos-2 0.933, TimesFM 0.932, TiRex 0.936, Bolt 1.054
- Global deep models at $h = 12$: DLinear 0.959, N-BEATS 0.999; Chronos-2 with cross-learning across the 27 countries: 0.845
- CRPS relative to AR (geometric mean over countries): Chronos-2 0.860, TimesFM 0.875, TiRex 0.865, ETS 0.972
- The 2021–2023 surge dominates every average: no univariate method foresaw it, so differences are about the speed of adaptation, not about foresight

Romania: forecasts from three origins



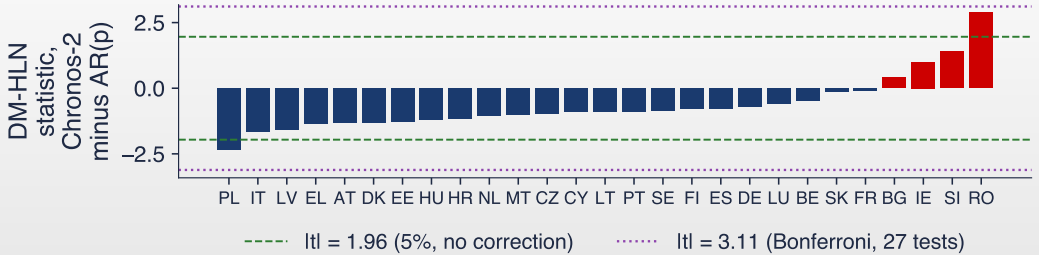
- Romanian HICP annual inflation and 12-month forecasts with 80% bands from Chronos-2 and AR(p), from June 2021, November 2022 and August 2025

ATS_ch13_benchmark

Interpreting the Romanian paths

- From June 2021: 12 months later inflation was 13.0%; Chronos-2 median 3.8% (80%: 1.5 – 6.6), AR 3.7%
- From November 2022, near the peak (14.6% in November 2022): outcome 6.9%; Chronos-2 16.5%, AR 13.7%
- From August 2025: outcome 6.3%; Chronos-2 11.4%, AR 9.0%; the latest observation is 6.3% (August 2026)
- Both methods extrapolate persistence; turning points come from information outside the series
  - energy prices, the end of the price caps, the VAT increase of August 2025 (its effect on annual inflation lasts exactly 12 months)
  - these are known in advance, hence natural covariates

DM tests for 27 countries



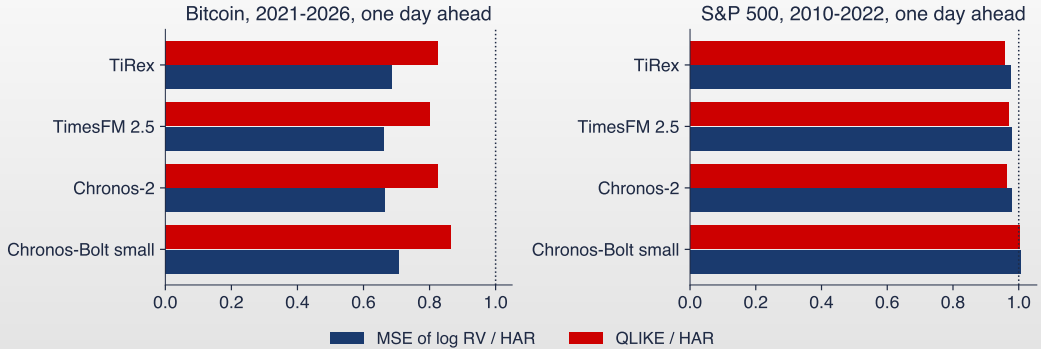
- Country-level DM–HLN statistics (HAC with $h-1$ lags) of the absolute errors of Chronos-2 minus AR(p) at $h = 12$; negative: Chronos-2 better

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Interpreting the multiple tests

- 23 of 27 statistics are negative; 2 are significant at 5% without correction (1 in favour of Chronos-2, 1 in favour of AR)
- With Holm: 0 rejections; with Benjamini–Hochberg: 0; Romania: $t = 2.88$, $p = 0.005$
- Pooled over the 27 countries (average loss differential at each origin, HAC variance): $t = -1.35$, $p = 0.180$
  - the bootstrap intervals of the previous chart treat countries as independent and look sharper than they are
- A paper that reports the countries where its model wins at 5% reports noise; the pooled test and the multiplicity-adjusted counts are the honest summary

Realised variance: foundation models against HAR



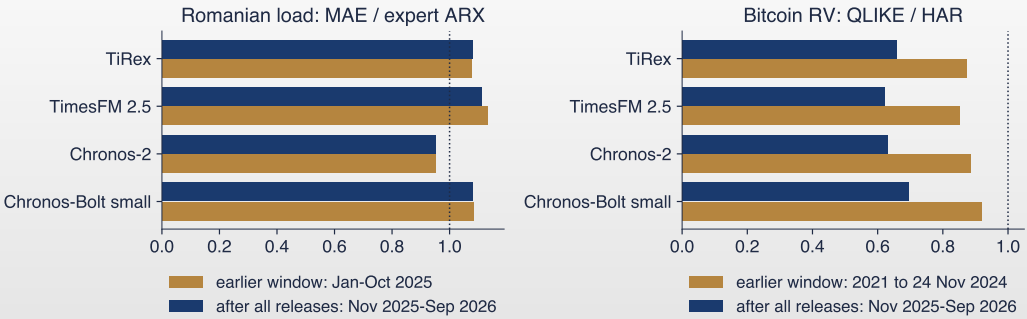
- One-day-ahead log RV: MSE and QLIKE relative to HAR; Bitcoin (Binance, 1 January 2021 – 18 September 2026, $n = 2\,084$), S&P 500 (Oxford-Man, 4 January 2010 – 25 February 2022, $n = 3\,047$)

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Interpreting the volatility comparison

- Bitcoin, QLIKE relative to HAR: Chronos-2 0.825, TimesFM 0.800, TiRex 0.824, Bolt 0.864; MCS (10%): {Chronos-2, TimesFM 2.5, TiRex}
- S&P 500: Chronos-2 0.963, TimesFM 0.968, TiRex 0.956, Bolt 1.004; MCS (10%): {Chronos-2, TimesFM 2.5, TiRex}
- Three of the four zero-shot models beat HAR on both assets, and HAR is outside both MCS
  - the slowly decaying memory of log RV (Chapter 10) is a shape a model can learn from a corpus
  - the gain is large for Bitcoin, small for the S&P 500
- Recent studies reach mixed verdicts Goel et al. (2025), Brini (2026), Rahimikia, Ni and Wang (2025)
  - gains depend on the asset, the horizon and the loss; fine-tuning or pretraining on financial data often matters

Accuracy before and after the model releases



- Relative loss of each model in an earlier window and in the window after every release (from 1 November 2025): load (MAE / expert ARX) and Bitcoin (QLIKE / HAR)

ATS_ch13_benchmark

Interpreting the contamination check

- Load, Chronos-2: 0.952 before, 0.950 after; TimesFM: 1.133, 1.112; the earlier window (January–October 2025) precedes both releases
- Bitcoin, Chronos-2: 0.886 (2021 – 24 November 2024) and 0.631 (after); TiRex: 0.874, 0.658; post-release windows have 334 and 322 days
- No deterioration after the releases (for Bitcoin the ratios are even lower): no evidence of contamination here
  - but the absence of a drop does not prove the absence of leakage, and markets change between windows

Recap: the evidence

- Strong shapes (load): the best foundation model matches or beats the domain model; weak shapes (returns): little to transfer
- Inflation: ahead of AR and ETS in the panel average (most with cross-learning), but no country difference survives Holm or BH and the pooled test is not significant
- Realised variance: zero-shot models beat HAR, clearly for Bitcoin, narrowly for the S&P 500
- Covariates in context help where the calendar matters; post-release windows show no sign of contamination

LLMTime: numbers as text

- Gruver et al. (2023): rescale, round to a fixed number of digits, write values as digits separated by spaces and commas; an off-the-shelf LLM continues the string
  - digit-by-digit probabilities define a hierarchical (multi-bin) density over continuous values; repeated sampling gives quantiles
- Findings of the paper
  - competitive zero-shot on some benchmarks
  - the tokenisation of numbers matters: a newer model could do worse because of how it splits digits
  - alignment (RLHF) hurts calibration
- Costs: thousands of tokens per series and many samples per forecast; the context window limits the history; the LLM may have memorised the data

Reprogramming a frozen LLM

- **Time-LLM** Jin et al. (2024)
  - patches of the series are mapped onto text prototypes (word embeddings)
  - a prompt prefix describes the task and the statistics of the series
  - the LLM stays frozen; a trained projection reads the forecast
- **One Fits All** (GPT4TS) Zhou et al. (2023)
  - a frozen GPT-2; only the input embedding, the layer norms and the output layer are trained
  - one model for forecasting, classification and anomaly detection
- The claim: knowledge from language transfers to series
  - the test of the claim is an ablation that removes the language model

Case study: the usefulness of language models for forecasting

- Tan et al. (2024): three LLM-based forecasters (including Time-LLM and GPT4TS), the same benchmarks, three ablations
- Ablations
  - remove the LLM; replace it by one attention layer; replace it by a basic Transformer block
- Result: no degradation, often an improvement
  - pretrained LLMs do no better than the same model trained from scratch
  - they do not model sequential dependence and do not help in few-shot settings
- Training and inference cost orders of magnitude more
- The method matters more than the result
  - an architecture claim needs an ablation with matched compute and tuning
- Scope of the critique
  - LLMs used as forecasters are affected
  - foundation models trained on series (Chronos, TimesFM, TiRex) are not
  - LLMs remain useful around forecasting: code, literature, critique (Section 9)

Memorisation and look-ahead

- LLMs recall their training period Lopez-Lira, Tang and Zhu (2025)
  - they reproduce macroeconomic and market values from that period with high precision, even when asked not to use future information
  - a backtest before the training cutoff measures recall, not forecasting
- The same logic for time-series foundation models
  - it applies to public series in the corpus (electricity, traffic, M4)
  - our evidence: windows after every release (Section 4)
  - the strongest evidence: live, pre-registered forecasts
- A practical rule
  - state the training cutoff of every model and the start of the evaluation window in the same table

Recap: LLMs

- LLMTime and reprogramming show that text models can be bent to series; ablations show the language part is rarely what helps
- Memorisation makes pre-cutoff evaluation of LLM forecasts invalid
- For forecasting, foundation models trained on series are the default; LLMs remain useful for code, literature and critique

Origins of conformal prediction



Alexander Gammerman, Royal Holloway, 2018

Photo: Jaguar224 (2018); CC BY-SA 4.0; Wikimedia Commons

- Vovk, Gammerman and Shafer (2005); Vovk, Gammerman and Shafer (2022)
  - prediction sets valid in finite samples
  - only exchangeability is assumed, for any predictive model
- The idea
  - how "conforming" a candidate value is to the data seen, measured by a rank
- Modern statistics Lei et al. (2018); Angelopoulos and Bates (2023)
  - adopted it as a wrapper around machine learning methods

Exchangeability and the target guarantee (1/2)

- Data: n calibration points and one test point, $Z_i = (X_i, Y_i)$, $i = 1, \ldots, n+1$
  - X_i: the predictors (features, lags, a model forecast); Y_i: the value to be covered; Z_{n+1}: the test point, whose Y_{n+1} is unknown
- **Exchangeability**: any reordering of the points has the same joint distribution

$$(Z_{\pi(1)}, \ldots, Z_{\pi(n+1)}) \stackrel{d}{=} (Z_1, \ldots, Z_{n+1}) \quad \text{for every permutation } \pi$$

  - π: a permutation (reordering) of $\{1, \ldots, n+1\}$; $\stackrel{d}{=}$: equal in distribution
- Examples
  - i.i.d. data are exchangeable
  - draws without replacement are exchangeable but dependent
  - a stationary AR(1) $y_t = \phi y_{t-1} + \varepsilon_t$, $0 < |\phi| < 1$, is not: $\operatorname{corr}(y_1, y_2) = \phi \neq \phi^2 = \operatorname{corr}(y_1, y_3)$, so swapping y_2 and y_3 changes the joint law

Exchangeability and the target guarantee (2/2)

- **Marginal coverage**: the prediction set contains the test value with probability at least $1 - \alpha$

$$\Pr\{Y_{n+1} \in \hat{C}(X_{n+1})\} \geq 1 - \alpha$$

  - $\hat{C}(x)$: the prediction set (usually an interval) built from the calibration data; α: the allowed miss rate, e.g. 0.1 for a 90% interval
  - the probability is taken over the calibration data and the test point together
- What it is not
  - **conditional** coverage, $\Pr\{Y \in \hat{C}(x) \mid X = x\} \geq 1 - \alpha$ for every value x
  - coverage given one particular calibration sample
- A **nonconformity score** $s(x, y)$: a number that is large when y is unusual given x
  - examples: $|y - \hat{\mu}(x)|$, $|y - \hat{\mu}(x)|/\hat{\sigma}(x)$, or the CQR score below
  - $\hat{\mu}(x)$: a point forecast of y given x; $\hat{\sigma}(x)$: a forecast of its spread (e.g. a GARCH volatility)

Split conformal prediction (1/2)

- Step 1: fit a forecasting model $\hat{\mu}$ on a training set Papadopoulos et al. (2002), Lei et al. (2018)
- Step 2: compute the scores of the n calibration points, which were not used in step 1

$$S_i = s(X_i, Y_i), \qquad i = 1, \ldots, n$$

- Step 3: the threshold $\hat{q}$ is the k-th smallest calibration score

$$\hat{q} = S_{(k)}, \qquad k = \lceil (n+1)(1-\alpha) \rceil$$

  - $S_{(k)}$: the k-th order statistic of $S_1, \ldots, S_n$; $\lceil \cdot \rceil$: rounding up to an integer; $\hat{q} = +\infty$ if $k > n$
  - example: $n = 99$, $\alpha = 0.1$ gives $k = 90$, the 90th smallest of the 99 scores
- Step 4: the prediction set keeps every candidate value whose score does not exceed the threshold

$$\hat{C}(x) = \{y : s(x, y) \leq \hat{q}\}$$

  - with $s(x, y) = |y - \hat{\mu}(x)|$: the interval $\hat{\mu}(x) \pm \hat{q}$

Split conformal prediction (2/2)

- **Theorem**: if the points (X_i, Y_i), $i = 1, \ldots, n+1$, are exchangeable, then

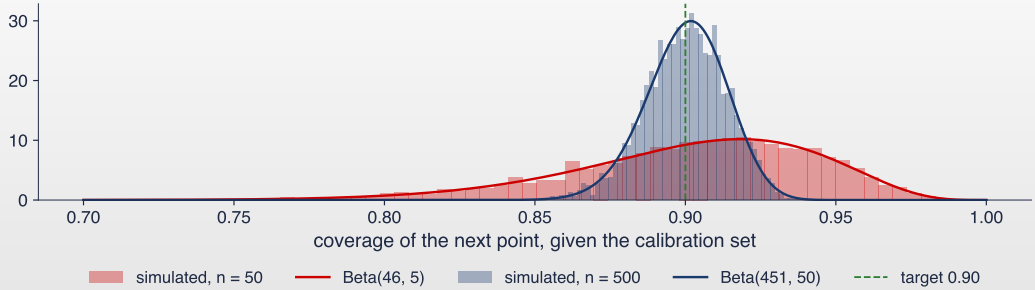
$$1 - \alpha \leq \Pr\{Y_{n+1} \in \hat{C}(X_{n+1})\} \leq 1 - \alpha + \frac{1}{n+1}$$

  - the upper bound requires no ties among the scores (continuous scores)

- Idea of the proof (Appendix)
  - given $\hat{\mu}$, the scores $S_1, \ldots, S_{n+1}$ are exchangeable, so the rank of S_{n+1} among them is uniform on $\{1, \ldots, n+1\}$
  - $Y_{n+1} \in \hat{C}(X_{n+1})$ exactly when this rank is at most k, which has probability $k/(n+1) \geq 1 - \alpha$

- No assumption on the model or on the distribution
  - a bad model gives wide intervals, not invalid ones

Coverage given the calibration set



- Coverage $F(\hat{q})$ of the next point for 4000 calibration sets of $n = 50$ and $n = 500$ scores, $\alpha = 0.1$, and the exact law Beta(k, $n + 1 - k$)
- F: the distribution function of the scores, so $F(\hat{q}) = \Pr\{S_{n+1} \leq \hat{q} \mid \text{calibration set}\}$ is the coverage obtained with one given calibration set

ATS_ch13_conformal_basics

Interpreting the coverage distribution

- Mean coverage 0.902 ($n = 50$) and 0.900 ($n = 500$), as the theory gives (0.902, 0.900): the marginal guarantee holds on average over calibration sets
- One calibration set is one draw: s.d. 0.041 and 0.014; with $n = 50$, 26% of the sets give coverage below 0.88, with $n = 500$ only 7%
- Since $F(\hat{q}) \sim \text{Beta}(k,\, n + 1 - k)$, choosing n is a sample-size calculation
  - the standard deviation of the realised coverage is about $\sqrt{\alpha(1-\alpha)/n}$: quadrupling n halves it

Conformalized quantile regression



Emmanuel Candès, 2012

Photo: Renate Schmid, MFO (2012); CC BY-SA 2.0 de; Wikimedia Commons

- Romano, Patterson and Candès (2019): fit two quantile regressions Koenker and Bassett (1978) on the training set
  - $\hat{q}_{\alpha/2}(x)$, $\hat{q}_{1-\alpha/2}(x)$: the estimated lower and upper conditional quantiles of Y given $X = x$
- **CQR score**: the signed distance of y outside the band

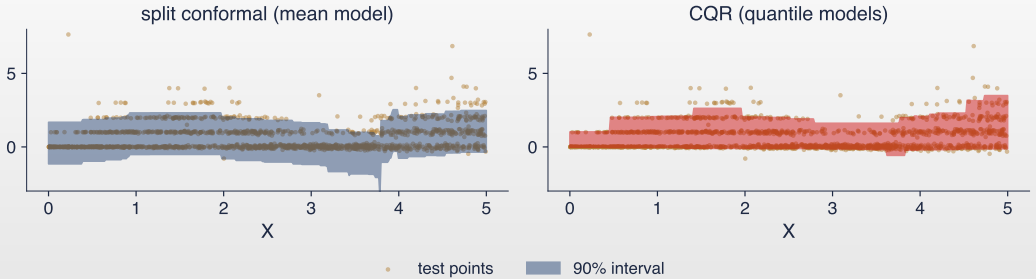
$$s(x, y) = \max\{\hat{q}_{\alpha/2}(x) - y,\ y - \hat{q}_{1-\alpha/2}(x)\}$$

  - negative inside the band, positive outside
- **Interval**: the band widened on both sides by the conformal threshold $\hat{q}$ of these scores

$$\hat{C}(x) = [\hat{q}_{\alpha/2}(x) - \hat{q},\ \hat{q}_{1-\alpha/2}(x) + \hat{q}]$$

  - $\hat{q} < 0$: the band was too wide and is narrowed
- Properties
  - marginal coverage by the split theorem; the width adapts to heteroskedasticity through the quantile models
  - the same score calibrates the quantiles of any foundation model (Section 8)

CQR on the design of Romano, Patterson and Candès



- Simulated data (their Figure 1): $Y = \text{Pois}(\sin^2 X + 0.1) + 0.03X\varepsilon_1 + 25 \cdot 1\{U < 0.01\}\varepsilon_2$, $X \sim U[0, 5]$
- $\text{Pois}(\lambda)$: a Poisson draw with mean λ; $\varepsilon_1, \varepsilon_2$: standard Normal noise; $U \sim U[0, 1]$, so 1% of the points get a large outlier; noise grows with X
- 1000 training and 1000 calibration points; gradient boosting for the mean and for the 5% and 95% quantiles

ATS_ch13_conformal_basics

Interpreting CQR

- Test coverage: split conformal 0.918, CQR 0.915, the raw quantile models 0.883; mean width 2.84, 2.20 and 2.17
- The raw quantile regressions undercover; CQR repairs them at almost no extra width, while the constant-width split band is 29% wider
- By X bins both stay within a few points of 90% (worst bins: CQR 0.892, split 0.896)
  - the gain of CQR is in width: narrow where Y is concentrated, wide where the Poisson noise is large

The limits of conditional coverage

- **Impossibility** Vovk (2012); Lei and Wasserman (2014); Foygel Barber et al. (2021)
  - if $\hat{C}$ has conditional coverage $\geq 1-\alpha$ at almost every x, for every distribution with a continuous X, its expected length is infinite
  - no finite-sample method can certify coverage at each point of a continuous covariate without assumptions
- What is achievable
  - coverage within each of a finite set of groups (Mondrian conformal: calibrate separately by group)
  - approximate conditional coverage under smoothness assumptions
  - coverage conditional on the calibration set, with high probability (Beta law)
- For time series the relevant condition is the past: the target is $\Pr\{Y_t \in \hat{C}_t \mid \mathcal{F}_{t-1}\} = 1-\alpha$
  - $\mathcal{F}_{t-1}$: the information available at $t-1$; $\hat{C}_t$: the set built at $t-1$ for period t
  - a VaR backtest (Chapter 9) tests exactly this property for one-sided sets

Recap: conformal prediction

- Split conformal: a rank argument gives finite-sample marginal coverage under exchangeability, for any model
- CQR adds adaptivity; the calibration sample size controls the variability of realised coverage
- Conditional coverage is impossible in general: test it, do not claim it

Conformal guarantees without exchangeability

- Time series violate exchangeability three ways: serial dependence, changing volatility, structural change; the last two break coverage most
- **Stationary and mixing**: split conformal remains approximately valid Oliveira et al. (2024)
  - the coverage gap shrinks with the calibration size and with the β-mixing coefficients, which measure how fast the dependence between distant observations dies out
  - block permutations restore exactness under weaker conditions Chernozhukov, Wüthrich and Zhu (2018)
- **Non-stationary**: no static method can work; the threshold must move with the data
  - weights (Barber et al.), refitting (EnbPI), feedback on errors (ACI, PID)
- **A common online frame** for the next slides
  - $S_t = s(X_t, Y_t)$: the score of period t, computed from an out-of-sample forecast
  - q_t: the threshold used at t, chosen from the past scores $S_1, \ldots, S_{t-1}$; the set is $\hat{C}_t = \{y : s(X_t, y) \leq q_t\}$
  - $\mathrm{err}_t = 1\{S_t > q_t\}$: the miss indicator, 1 if Y_t falls outside $\hat{C}_t$

Weighted conformal prediction (1/2)

- Barber et al. (2023): each calibration score gets a fixed weight $w_i \in [0, 1]$, chosen before seeing the data; the test point gets weight $w_{n+1} = 1$
  - normalised weights $\tilde{w}_i = w_i / (1 + \sum_{j=1}^{n} w_j)$, $i = 1, \ldots, n+1$; they sum to 1
- The threshold $\hat{q}$ is the $(1 - \alpha)$ quantile of the weighted distribution of the scores, with the test point placed at $+\infty$

$$\hat{q} = Q_{1-\alpha}\Big(\sum_{i=1}^{n} \tilde{w}_i \, \delta_{S_i} + \tilde{w}_{n+1} \, \delta_{+\infty} \Big)$$

  - δ_a: a unit point mass at a; $Q_{1-\alpha}(\cdot)$: the $(1 - \alpha)$ quantile of a distribution
  - with all $w_i = 1$ this is the split conformal threshold
- **Covariate shift** with a known likelihood ratio: the weights $w(x) = dP_{\text{test}}/dP_{\text{train}}$ give exact coverage Tibshirani et al. (2019)
  - $dP_{\text{test}}/dP_{\text{train}}$: the ratio of the densities of X in the test and in the training population

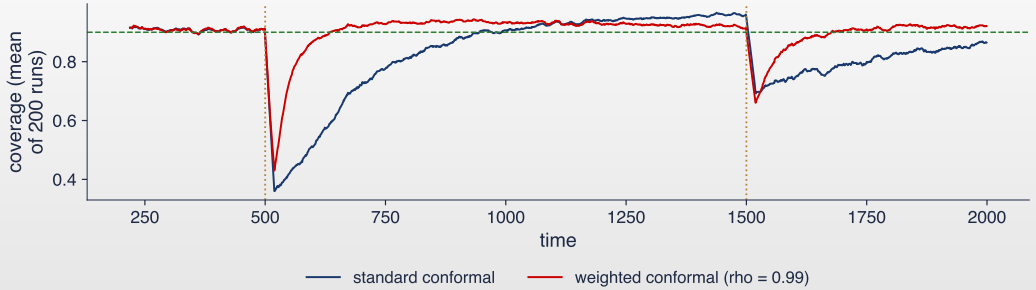
Weighted conformal prediction (2/2)

- **Theorem** Barber et al. (2023): the coverage loss is bounded by the weighted distance from exchangeability

$$\Pr\{Y_{n+1} \in \hat{C}(X_{n+1})\} \geq 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \, d_{\mathrm{TV}}(Z, Z^i)$$

  - $Z = (Z_1, \ldots, Z_{n+1})$: the data; Z^i: the same data with the test point and point i swapped
  - d_{TV}: the total variation distance between two distributions, in $[0, 1]$; 0 when they coincide
- Under exchangeability every $d_{\mathrm{TV}}(Z, Z^i) = 0$ and the gap vanishes
- Under slow drift, d_{TV} is small for recent points: give them the largest weights
  - geometric weights $w_i = \rho^{n+1-i}$, $\rho \in (0, 1)$: the weight halves every $\ln 2 / |\ln \rho|$ steps
  - effective sample size $\approx 1/(1-\rho)$: $\rho = 0.99$ uses about 100 recent scores
  - a smaller ρ adapts faster but makes the threshold more variable

Weighted conformal under changepoints



- Simulated regression $Y_t = X_t'\beta_t + \varepsilon_t$: $X_t \sim N(0, I_4)$, four independent standard Normal regressors (I_4: the 4×4 identity matrix); ε_t: Normal noise; the coefficient vector β_t changes at $t = 500$ and $t = 1500$
- Least squares on all past data; scores: absolute one-step-ahead residuals (prequential: each computed before Y_t is used); coverage averaged over 200 runs (20-step moving average)

ATS_ch13_conformal_time

Interpreting the weighted method

- Average coverage after the burn-in: standard 0.836, weighted ($\rho = 0.99$) 0.900; worst 20-step average 0.36 against 0.43
- Both collapse at a break, because the least-squares fit is wrong for a while; the weighted threshold forgets the old scores and recovers within about 100 steps
- Price: mean width 7.47 against 6.22; the guarantee is a bound in terms of d_{TV}, not exact coverage

EnbPI: ensembles without data splitting

- Xu and Xie (2021), Xu and Xie (2023): fit B models on B bootstrap samples of the training period (blocks of consecutive observations, to respect dependence)
  - leave-one-out residuals: for point i, aggregate only the models whose bootstrap sample excluded i
- Interval at t: the ensemble forecast plus or minus a quantile of recent absolute residuals

$$\hat{C}_t = \hat{f}(x_t) \pm Q_{1-\alpha}(|e_{t-W}|, \ldots, |e_{t-1}|)$$

  - $\hat{f}(x_t)$: the aggregated forecast of the B models given the predictors x_t; e_s: the residual of period s; W: the window length
  - after y_t is observed, its residual enters the window and the oldest one leaves
- No refitting at each step, no calibration split: efficient for long streams
- Guarantee: asymptotic, conditional coverage if the errors are stationary and strongly mixing and the ensemble is consistent; designed for energy series (solar and wind)

Adaptive conformal inference (1/2)

- Gibbs and Candès (2021): the nominal level is replaced by a working level α_t that is updated after every observation
  - q_t: the conformal $(1-\alpha_t)$ quantile of the recent scores; α: the target miss rate (e.g. 0.1)
- **Update**: after observing Y_t, move the working level against the last error

$$\alpha_{t+1} = \alpha_t + \gamma\,(\alpha - \mathrm{err}_t)$$

  - $\gamma > 0$: the step size; $\mathrm{err}_t \in \{0, 1\}$: the miss indicator of the previous slides
- Reading the update
  - a miss ($\mathrm{err}_t = 1$) lowers α_t by $\gamma(1-\alpha)$: the next set is wider
  - a hit ($\mathrm{err}_t = 0$) raises α_t by $\gamma\alpha$: the next set is narrower
  - $\alpha_t \leq 0$ gives $q_t = +\infty$ (the whole real line); $\alpha_t \geq 1$ gives the empty set

Adaptive conformal inference (2/2)

- **Theorem** Gibbs and Candès (2021): for any sequence of data, the miss rate over T periods is close to α

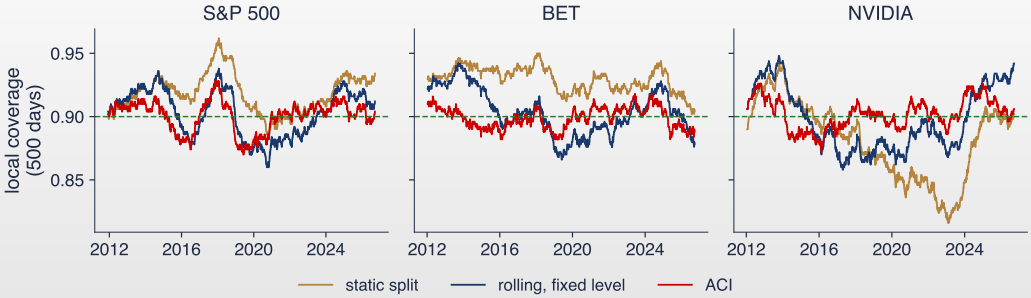
$$\left|\frac{1}{T}\sum_{t=1}^{T}\mathrm{err}_t - \alpha\right| \leq \frac{\max\{\alpha_1, 1-\alpha_1\} + \gamma}{\gamma T}$$

  - α_1: the starting level; the bound falls as $1/T$, and is tighter for a larger γ

- Idea of the proof (▸ Appendix)
  - α_t always stays in $[-\gamma, 1+\gamma]$
  - summing the updates, $\alpha_{T+1} - \alpha_1 = \gamma\sum_t(\alpha - \mathrm{err}_t)$; divide by γT

- A long-run frequency, not conditional coverage
  - the guarantee holds even for an adversarially chosen sequence
  - choosing γ adaptively: AgACI Zaffran et al. (2022), DtACI Gibbs and Candès (2024); ACI for VaR: Chapter 9

ACI on stock-market volatility



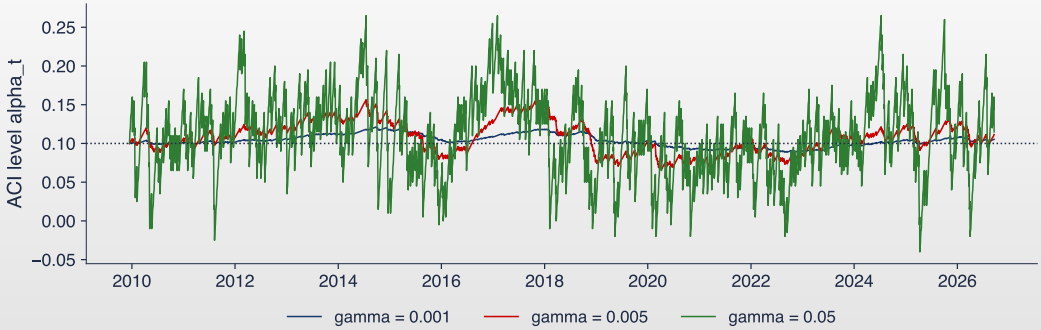
- Design of Gibbs and Candès (2021): the target is $V_t = r_t^2$, the squared daily return; $\hat{\sigma}_t^2$: its GARCH(1,1) forecast, fitted on the previous 1250 days
- Score $|V_t - \hat{\sigma}_t^2|/\hat{\sigma}_t^2$ (relative error of the variance forecast), $\alpha = 0.1$, $\gamma = 0.005$; local coverage over 500 days; evaluation from 15 December 2009

ATS_ch13_conformal_time

Interpreting ACI on volatility

- Three thresholds compared
  - static: split conformal calibrated once, on the first 1250 scores
  - rolling: split conformal on the last 1250 scores, with the fixed level α
  - ACI: the same rolling window, with the level α_t
- Overall coverage, S&P 500: static 0.918, rolling 0.903, ACI 0.901; BET: 0.928, 0.903, 0.899; NVIDIA: 0.882, 0.900, 0.901
- Local coverage of the static method ranges from 0.884 to 0.962 (S&P 500); ACI stays within 0.870 – 0.928
- Christoffersen conditional coverage of the misses (S&P 500): static $p < 0.001$, ACI $p = 0.968$
  - a calibration fixed on 2005–2009 is too wide afterwards
  - updating restores both the frequency and the independence of misses

The step size of ACI



⊡ S&P 500 scores of the previous chart: the level α_t for $\gamma = 0.001$, 0.005 and 0.05

ATS_ch13_conformal_time

Interpreting the step size

- Miss rates 0.0987, 0.0994, 0.0999 against the bounds 0.2138, 0.0430, 0.0045 ($T = 4214$): the theorem holds with room to spare
- Large γ: α_t ranges -0.04 – 0.27, infinite intervals on 1.6% of days; small γ: slow reaction, long runs of misses after a shock
- γ trades adaptivity against stability: tune it on a past window, or let DtACI aggregate several values

Conformal PID control (1/2)

- Angelopoulos, Candès and Tibshirani (2023), **quantile tracking** (the P term): the threshold itself is updated after every observation

$$q_{t+1} = q_t + \eta\,(\mathrm{err}_t - \alpha)$$

  - $\eta > 0$: the step size, in the units of the scores; a miss raises the threshold by $\eta(1-\alpha)$, a hit lowers it by $\eta\alpha$
  - this is online gradient descent on the pinball loss of the score at level $1-\alpha$
- **Proposition**: if the scores lie in $[0, B]$, the miss rate is close to α for any sequence

$$\Big|\frac{1}{T}\sum_{t=1}^{T}(\mathrm{err}_t - \alpha)\Big| \le \frac{B+\eta}{\eta T}$$

  - B: an upper bound of the scores; the threshold moves on the scale of the scores, not of α as in ACI
- ACI is the special case that tracks the level instead of the quantile

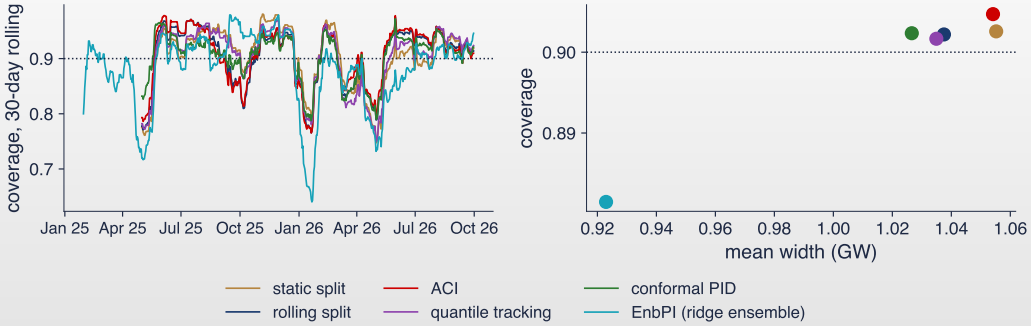
Conformal PID control (2/2)

- **Integrator** (the I term): adds a correction driven by the accumulated coverage error

$$r_t\Big(\sum_{i\le t}(\mathrm{err}_i-\alpha)\Big), \qquad r_t(x)=K_I\tan\left(x\log t/(tC_{\mathrm{sat}})\right)$$

  - $\sum_{i\le t}(\mathrm{err}_i-\alpha)$: misses in excess of the target, accumulated up to t
  - $K_I>0$: the gain of the integrator; $C_{\mathrm{sat}}>0$: the saturation constant; the tan grows without bound near its pole, which keeps the long-run guarantee
- **Scorecaster** (a D-like term): a model that forecasts the next score from its past and is added to the threshold
  - it anticipates predictable patterns in the errors: seasonality, trends
- The name comes from control theory
  - proportional (react to the last error), integral (to the accumulated error), derivative (to the expected change)

Online methods on Romanian load



- 90% day-ahead intervals around the Chronos-Bolt median, one score stream per hour (24 streams), calibration window 91 days; EnbPI on the expert ARX regressors (ridge, 20 block-bootstrap models); evaluation from 2 April 2025

ATS_ch13_conformal_time

Interpreting the online methods

- Coverage: static 0.903, rolling 0.902, ACI 0.905, quantile tracking 0.902, PID 0.902, EnbPI 0.881
- Mean width (GW): 1.06, 1.04, 1.05, 1.03, 1.03, 0.92; worst 30-day coverage: static 0.747, PID 0.787
- Over the whole period every method except EnbPI is close to 90%; the differences are in the dips after seasonal transitions and in width, where PID is best on both counts
- EnbPI centres the band on its own ridge ensemble, less accurate than the base forecast: narrower intervals, lower coverage

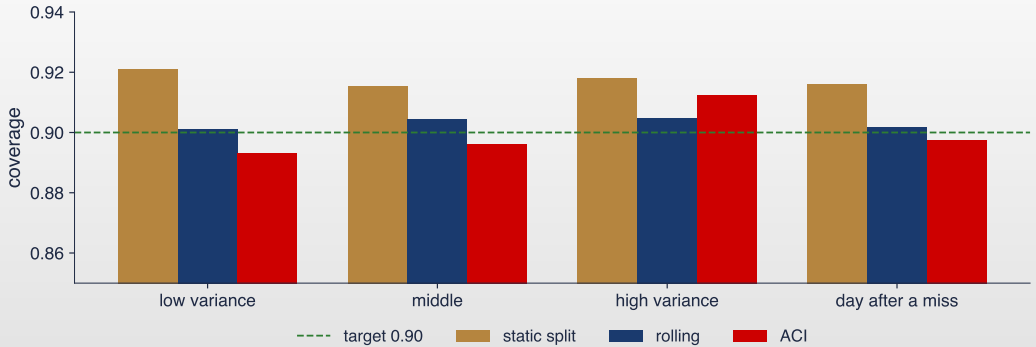
Coverage diagnostics

- **Frequency and independence** of the misses (Chapter 9)
  - miss rate and the Kupiec test Kupiec (1995) (is the rate equal to α?); Christoffersen test Christoffersen (1998) (are misses independent over time?)
  - regression check in the spirit of the DQ test Engle and Manganelli (2004): regress $\mathrm{err}_t - \alpha$ on lagged misses and on the predicted width
- **Local coverage**: the empirical counterpart of conditional coverage
  - coverage over rolling windows
  - coverage by classes of a variable known at the origin (predicted volatility, hour, regime)
- **Sharpness**: the mean width, and the **interval score** Winkler (1972), Gneiting and Raftery (2007), which adds a penalty for every miss

$$\mathrm{IS} = (u - \ell) + \frac{2}{\alpha}(\ell - y)1\{y < \ell\} + \frac{2}{\alpha}(y - u)1\{y > u\}$$

  - $[\ell, u]$: the central $(1 - \alpha)$ interval; $u - \ell$: its width; y: the outcome; lower is better
  - a proper scoring rule: it rewards narrow intervals only if they keep the nominal coverage
- Report all three: a method that is valid but twice as wide is not a better method

Coverage by volatility regime



- S&P 500, 90% volatility intervals of the ACI chart: coverage by tercile of the GARCH variance forecast and on the day after a miss

ATS_ch13_conformal_time

Interpreting conditional coverage

- Static: 0.921 (low variance), 0.918 (high), 0.916 after a miss; ACI: 0.893, 0.912, 0.897
- Coverage is almost flat across volatility terciles and after a miss, for every method: dividing by the GARCH variance makes the score close to pivotal
- Contrast: with the raw score $|r_t|$ (Seminar 13, B3) misses cluster in turbulent years whatever the calibration; conditional coverage is a property of the score, not of the conformal step

Recap: conformal methods for time series

- Weighted conformal: a coverage bound in total variation; geometric weights adapt to drift
- EnbPI: bootstrap ensembles and a sliding residual window, asymptotic validity under mixing
- ACI and PID: feedback on misses gives long-run coverage for any sequence; conditional coverage must still be tested

The need to calibrate foundation-model quantiles (1/2)

- Pretrained quantiles are calibrated on the corpus distribution, not on your series
  - miscalibration is the default, in either direction (bands too narrow or too wide)
- Most models output only deciles (Section 2): a 95% interval or a VaR 1% is beyond their trained levels
- Procedure: online CQR on the model's own 10%–90% band

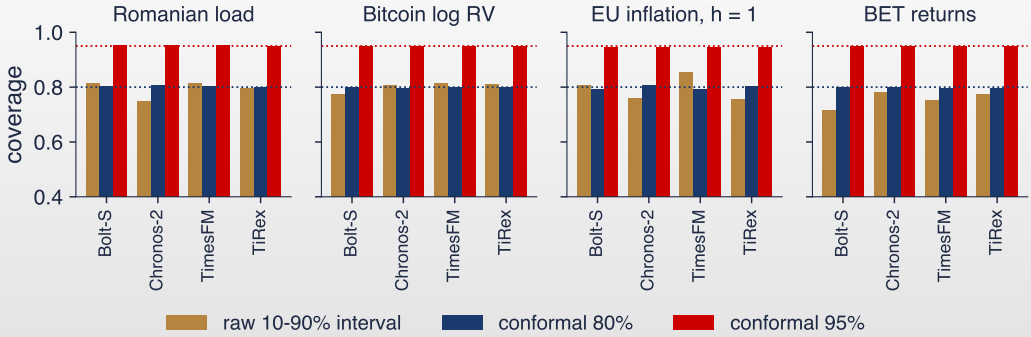
$$S_t = \max\{\hat{q}_{0.1,t} - y_t,\ y_t - \hat{q}_{0.9,t}\}, \qquad \hat{C}_t = [\hat{q}_{0.1,t} - q_t,\ \hat{q}_{0.9,t} + q_t]$$

  - $\hat{q}_{0.1,t}$, $\hat{q}_{0.9,t}$: the 10% and 90% quantiles forecast by the model for period t; S_t: the CQR score
  - q_t: the threshold chosen by ACI for any target level (80% or 95%)

The need to calibrate foundation-model quantiles (2/2)

- One-sided version for VaR
  - score $S_t = \hat{q}_{0.1,t} - y_t$; threshold q_t tracked at level α
  - the calibrated quantile is $\hat{q}_{0.1,t} - q_t$
  - $\mathrm{VaR}_t = -(\hat{q}_{0.1,t} - q_t)$: VaR expressed as a positive loss
- What the shift changes
  - $q_t > 0$ lowers the quantile (a larger VaR); $q_t < 0$ raises it
  - the model itself is not re-estimated: only its output is recalibrated
- Further reading
  - conformal VaR recalibration of foundation models: Pele et al., Conformal_Oracle (GitHub); Pele et al., TSFM_VaR_CEE (GitHub)

Raw and conformal coverage of foundation models



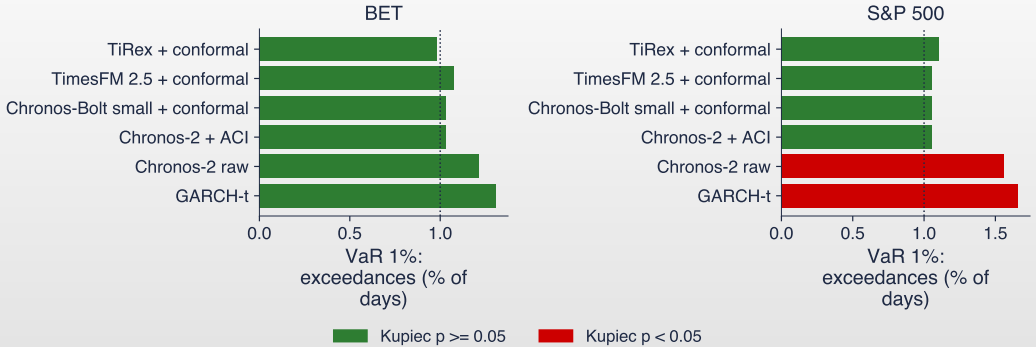
Coverage of the raw 10–90% band and of the online-CQR 80% and 95% intervals (ACI, window 250, $\gamma = 0.005$): Romanian load (24 hourly streams), Bitcoin log RV, EU inflation at $h = 1$ (27 streams), BET daily returns

ATS_ch13_calibration

Interpreting the calibration

- Raw 80% bands: load 75–81%, Bitcoin 77–81%, inflation 76–85%, BET returns 72–78% across the four models
- After online CQR: 80% intervals cover 79–81%, 95% intervals 94.5–95.4% in every task
- Raw bands are miscalibrated in both directions: too narrow (Chronos-2 on load), too wide (TimesFM on inflation)
  - the conformal step widens or narrows them: mean width between 0.90 and 1.24 times the raw width
  - and it reaches 95% from models that output only deciles

Value at Risk from foundation models



- VaR 1% exceedances, BET and S&P 500 from 28 December 2017 (2 175 days): GARCH-t (rolling 1000 days, Chapter 9), Chronos-2 raw and with ACI, deciles of the other models extended by a one-sided conformal shift

ATS_ch13_calibration

Interpreting the VaR backtest

- BET, VaR 1%: GARCH-t 1.31% (Kupiec $p = 0.170$), Chronos-2 raw 1.22% ($p = 0.331$), Chronos-2 + ACI 1.03% ($p = 0.893$), TimesFM + conformal 1.08%
- S&P 500: GARCH-t 1.66% (Kupiec $p = 0.007$), Chronos-2 raw 1.56% ($p = 0.021$), + ACI 1.06% ($p = 0.802$, Christoffersen $p = 0.454$)
- Calibration fixes the exceedance rate of every model, but it does not add information about the tail
  - the conformally extended deciles have a larger quantile loss than GARCH-t (BET, Chronos-Bolt: DM $t = 2.70$)
  - Chronos-2, trained on 1% quantiles, does not ($t = 0.10$)

Recap: calibration

- Treat foundation-model quantiles as scores to be calibrated, not as probabilities
- Online CQR with ACI gives target coverage at any level, including beyond the trained quantiles
- Backtest calibrated VaR as in Chapter 9: frequency, independence and loss

An open question

- The question
  - do time-series foundation models beat domain-specific econometric models on Central and Eastern European data?
  - in a window after their release dates, and once multiplicity is accounted for
- A testable form
  - formal: H_0: equal expected loss (pooled over pre-registered series) of the best foundation model and the best domain model, in a window that starts after every model release
  - falsified by a pooled DM test at 5% and an MCS that excludes one side, on series and horizons fixed before the data arrive
- Why it matters
  - central banks and grid operators consider replacing tuned models by zero-shot ones
  - published wins often come from a favourable choice of series
  - literature to start from: Aksu et al. (2024), Shchur et al. (2025), Meyer et al. (2025), Pan and Ezzat (2026), Brini (2026)

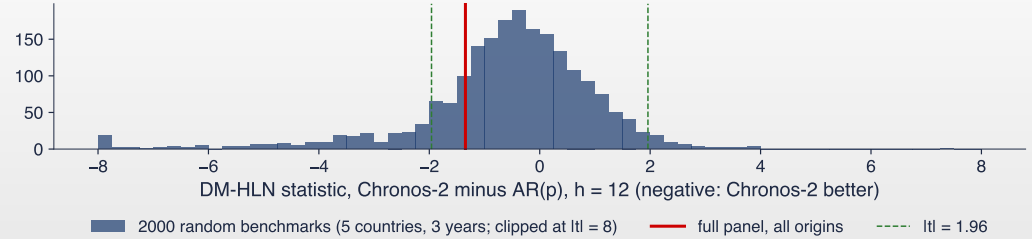
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed or arXiv papers since 2024 that evaluate time-series foundation models only after their release dates; give arXiv IDs or DOIs.` Then check every identifier
  - **hypothesis**: `Write a pre-registration: series, origins, horizons, losses, baselines, the pooled test and the MCS level.`
  - **code and replication**: ask for a wrapper that returns quantiles at fixed levels for each model; replicate a known number first (the Chronos-Bolt quantile range, our load MAE)
  - **critique**: `Act as a hostile referee: list every way this benchmark could favour the foundation models.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (arXiv ID or DOI resolves, the title matches, the result is in the paper)
- Release dates and training cutoffs of every model, from the model cards, against the start of the evaluation window
- Quantile levels actually produced by each model (a request for 1% may silently return 10%)
- Baselines tuned with the same care; the context available to every model is the same
- Multiplicity: number of series, horizons and models tested; the pooled test and adjusted p-values reported

Mini-case: sensitivity of the comparison to the choice of benchmark



- DM–HLN statistics of Chronos-2 against AR(p) at $h = 12$ on 2000 random benchmarks of 5 EU countries and 3 years of origins, and on the full panel
- 11.2% of the random benchmarks declare Chronos-2 significantly better and 2.1% significantly worse; the full panel gives $t = -1.35$ ($p = 0.180$)
- An AI summary that presents one such paper as evidence for (or against) foundation models is wrong; only the pre-registered panel answers the question

ATS_ch13_benchmark

Project idea

- **A pre-registered, post-release benchmark of foundation models for CEE macro and risk, with conformal calibration**
  - replicate: the zero-shot protocol of Ansari et al. (2024) (WQL, MASE, geometric means) and the ACI volatility design of Gibbs and Candès (2021) on our data
  - extend: Romanian, Polish, Hungarian and Czech inflation, load and stock indices; covariates in context (energy prices, calendars); online CQR to 95% and VaR 1%; pooled tests with Holm and BH
  - pre-register: series, origins after 1 November 2025, horizons, losses, baselines and the decision rule; forecasts for the next months written down before the data arrive
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Foundation models amortise estimation over a corpus; design choices (scaling, patches, quantile range) set what they can and cannot forecast
- On our data the best zero-shot model is competitive with domain models; gains depend on the task and do not survive cherry-picking
- Evaluate after the release, pool across series, correct for multiplicity
- Conformal prediction gives finite-sample marginal coverage under exchangeability; under dependence, weights, ensembles and feedback restore long-run coverage
- Conditional coverage is impossible in general: diagnose it with Christoffersen tests and coverage by regime

Self-assessment

Questions

- Why can Chronos not forecast a value above $15s$?
- Why does a stationary AR(1) violate exchangeability?
- What is the law of the coverage of split conformal given the calibration set?
- Which quantity stays bounded in the proof of the ACI theorem?
- Why is a post-release window necessary in a foundation-model benchmark?

Next:

- Causal inference for time series
- Granger and structural causality, causal discovery, policy evaluation

Appendix: proof of split conformal coverage

- Condition on the training set, so $\hat{\mu}$ and s are fixed; exchangeability of $Z_1, \ldots, Z_{n+1}$ implies exchangeability of $S_1, \ldots, S_{n+1}$
- With no ties, the rank R of S_{n+1} among $S_1, \ldots, S_{n+1}$ is uniform on $\{1, \ldots, n+1\}$ (every ordering is equally likely)
- $S_{n+1} \leq S_{(k)}$ (order statistic of the first n) $\iff$ $R \leq k$; so $\Pr\{Y_{n+1} \in \hat{C}\} = k/(n+1)$
- $k = \lceil (n+1)(1-\alpha) \rceil$ gives $1 - \alpha \leq k/(n+1) < 1 - \alpha + 1/(n+1)$; ties only increase coverage
- Given the calibration set, coverage is $F(S_{(k)})$ with F the score distribution; $F(S_i)$ are i.i.d. uniform, so $F(S_{(k)}) \sim \text{Beta}(k,\ n+1-k)$

Back

Appendix: the two online guarantees

- **ACI**: if $\alpha_t < 0$ then $q_t = +\infty$ and $\mathrm{err}_t = 0$, so $\alpha_{t+1} = \alpha_t + \gamma\alpha > \alpha_t$; if $\alpha_t > 1$, $\mathrm{err}_t = 1$ and α_t falls
  - hence $\alpha_t \in [-\gamma, 1+\gamma]$ for all t; summing the updates, $\gamma\left|\sum_{t\le T}(\alpha - \mathrm{err}_t)\right| = |\alpha_{T+1} - \alpha_1| \le \max\{\alpha_1, 1-\alpha_1\} + \gamma$
- **Quantile tracking**: if $q_t > B$ then $\mathrm{err}_t = 0$ and q_t falls by $\eta\alpha$; if $q_t < 0$ then $\mathrm{err}_t = 1$ and q_t rises
  - hence $q_t \in [-\eta, B+\eta]$ (starting inside); $\eta\sum_{t\le T}(\mathrm{err}_t - \alpha) = q_{T+1} - q_1$, so the average gap is at most $(B+\eta)/(\eta T)$
- Neither proof uses any probability: the guarantees hold for every sequence, which is why they say nothing about conditional coverage

Back

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